ESC European Society of Cardiology

European Heart Journal: Acute Cardiovascular Care (2025) **14**, 711–719
https://doi.org/10.1093/ehjacc/zuaf110

ORIGINAL SCIENTIFIC PAPER
Thromboembolic Diseases

Endovascular mechanical thrombectomy vs. catheter-directed thrombolysis in pulmonary embolism: insights from the National Readmission Database

Monil Majmundar [1], Vidit Majmundar[2], Vivek Bhat [3], Kunal N. Patel[1], Gaurav Parmar[4], Adam Alli[5], Aaron Rohr[5], Peter Monteleone[6], Sanjum S. Sethi [7], and Kamal Gupta [1]*

[1]Department of Cardiovascular Medicine, University of Kansas Medical Center, 3901 Rainbow Blvd 27 Mailstop 4023, Kansas City, KS 66160, USA; [2]Department of Internal Medicine, Saint Vincent Hospital, Worcester, MA, USA; [3]Department of Medicine, SUNY Upstate Medical University, Syracuse, NY, USA; [4]Section of Vascular Medicine, Massachusetts General Hospital, Boston, MA, USA; [5]Department of Radiology, University of Kansas Medical Center, Kansas City, KS, USA; [6]Division of Cardiovascular Medicine, University of Texas at Austin, Dell Medical School, Austin, TX, USA; and [7]Division of Cardiology, Columbia University Irving Medical Center, New York, NY, USA

Received 8 December 2024; revised 17 February 2025; accepted 6 August 2025; online publish-ahead-of-print 23 August 2025

Aims

Catheter-based therapies for pulmonary embolism (PE), including endovascular mechanical thrombectomy (MT) and catheter-directed thrombolysis (CDT), are increasingly being used in clinical practice. However, real-world comparative data between these two modalities are scarce. We aimed to evaluate and compare the outcomes of MT and CDT in patients with PE.

Methods and results

This retrospective cohort study utilized the 2021 National Readmission Database (NRD) to identify adults with a primary diagnosis of PE who underwent either MT or CDT. The primary outcome was in-hospital mortality, while secondary outcomes included major bleeding, cardiac arrest, vascular complications, and post-discharge readmissions. Propensity-score matching was applied, followed by logistic and Cox-proportional hazard regression analyses. Subgroup analyses were conducted based on hospital procedural volume. After propensity-score matching, 7376 patients who underwent MT and 7355 who underwent CDT were included. MT was associated with higher odds of in-hospital mortality (4.4% vs. 3.4%; OR: 1.31, 95% CI: 1.01–1.68; $P = 0.04$) and major bleeding (6.3% vs. 3.6%; OR: 1.79, 95% CI: 1.39–2.32; $P < 0.001$) compared with CDT. No significant differences were observed in post-discharge mortality, although all-cause readmissions were higher in the MT group. Higher hospital procedural volumes were associated with lower in-hospital mortality and lower major bleeding rates in both MT and CDT.

Conclusion

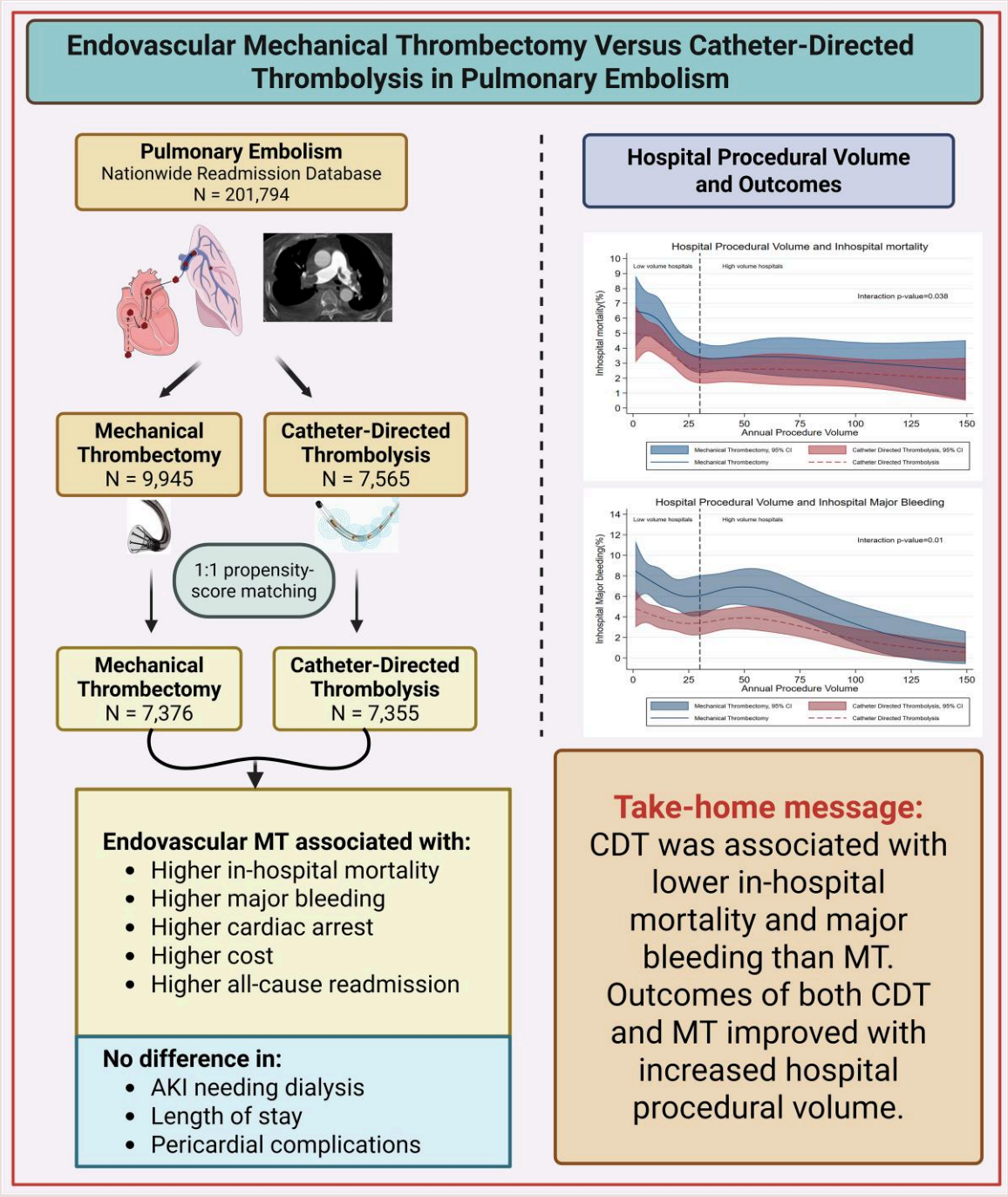
Endovascular CDT was associated with lower in-hospital mortality and major bleeding compared to MT in PE. As hospital procedural volume increased, both these outcomes improved, while difference in outcomes between MT and CDT reduced.

* Corresponding author. Tel: +1 (913) 588-1227, Email: kgupta@kumc.edu

© The Author(s) 2025. Published by Oxford University Press on behalf of the European Society of Cardiology. All rights reserved. For commercial re-use, please contact reprints@oup.com for reprints and translation rights for reprints. All other permissions can be obtained through our RightsLink service via the Permissions link on the article page on our site—for further information please contact journals.permissions@oup.com.

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

Graphical Abstract



Keywords Catheter-based therapy • Major bleeding • Thrombolysis • Venous thromboembolism

Introduction

Pulmonary embolism (PE) is the third leading cause of cardiovascular death,[1] and its incidence has been increasing over the last few decades.[2] Reperfusion therapy is often warranted in high-risk and selected intermediate-risk PE cases to rapidly alleviate right ventricular strain and improve haemodynamics. Systemic thrombolysis (ST) has long been considered the first-line treatment for haemodynamically unstable PE.[3,4] However, despite its efficacy, ST carries a high risk of major bleeding, notably intracranial haemorrhage, which has limited its widespread use, particularly in elderly patients and those with contraindications.[5] As a result, there has been a growing interest in catheter-based

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

interventions, such as catheter-directed thrombolysis (CDT) and mechanical thrombectomy (MT), which offer the potential for effective clot removal or dissolution with possibly lower bleeding risk.

CDT employs the delivery of thrombolytic agents directly into the affected pulmonary artery via a multi-hole catheter. A commonly used version of the catheter utilizes ultrasound assistance to enhance thrombolysis and was approved by the US FDA in 2014. This technique has demonstrated promising outcomes in smaller studies and single-arm registries, such as the SEATTLE II study, a single-arm registry that found that ultrasound-facilitated CDT effectively reduced right ventricular strain, and the ULTIMA study, which showed that CDT significantly reduced right ventricular strain compared to heparin alone at 24 h. While these studies did not compare CDT directly with systemic thrombolysis, there appeared to be a lower incidence of major bleeding than prior studies utilizing systemic thrombolysis.[6,7] Similarly, MT has emerged as an alternative approach for treatment of intermediate-risk PE, employing mechanical devices to aspirate or fragment the thrombus, with multiple single-arm studies such as FLASH, FLARE, EXTRACT-PE, and STRIKE-PE demonstrating favourable safety and efficacy profiles.[8–11]

Despite the encouraging treatment results with both CDT and MT, there remains a paucity of data directly comparing these two modalities in patients with PE, and much of the current decision-making is based on clinician or institutional preference. Previous retrospective analyses have often focused on single-centre experiences (generally high-volume centres) or specific device types,[7–11] limiting the generalizability of their findings. Additionally, few studies have adequately adjusted for potential confounders or assessed outcomes in centres with lower case volume, which could significantly impact procedural success and complication rates. The recently published PEERLESS randomized controlled trial (RCT) study added significantly to the evidence base by directly comparing MDT with CDT in intermediate-risk PE patients. However, it is not known if these results are representative of a real-world clinical practice, where these procedures are performed in a variety of hospital settings and in a much broader variety of patients.[12]

Given this knowledge gap, we aimed to comprehensively compare endovascular MT and CDT in all patients with PE. We attempted this by analysing a large, nationally representative database, using propensity-score matching to minimize bias.

Methods

The study population was derived from the National Readmission Database (NRD) 2021. NRD is an all-payer inpatient database sponsored by the Healthcare Cost and Utilization Project (HCUP), established by the Agency for Healthcare Research and Quality.[13] It includes data from hospitals in 28 geographically dispersed states in the USA, representing ~58.7% of all US hospitalizations. Each patient is tracked using a unique de-identified number to determine hospital readmissions within a calendar year.

This study was deemed exempt from the Kansas University Medical Center institutional review board, and informed consent was not required, as the NRD contains de-identified datasets with prior ethical committee approval. This study followed the reporting guidelines specified by the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement (see Supplementary material online, *Table S1*).

Study population

All adults (aged >18 years) with a primary diagnosis of PE, who underwent endovascular CDT or MT were included. For this, we queried the NRD using the International Classification of Diseases, Tenth Revision, Procedure Coding System and Clinical Modification (ICD-10-PCS, ICD-10-CM). These codes were similar to those in prior studies, and have been summarized in Supplementary material online, *Table S2*.[14,15] We excluded patients who underwent both CDT and MT, surgical thrombectomy, and those who received ST. Included patients were then subjected to 1:1 propensity-score matching. *Figure 1* summarizes the patient inclusion process.

Further stratification was conducted to compare outcomes of patients with haemodynamically significant PE, defined as those with saddle PE or PE with cor pulmonale, and those undergoing intervention < 24 h after presentation.

Patient and hospital characteristics

We collected data on patients' age and gender. To account for underlying health conditions, we adjusted for patients' Elixhauser comorbidity index.[16] Additionally, we used ICD-10-CM codes to identify prior comorbidities, including coronary artery disease (CAD), stroke/transient ischaemic attack (TIA), cancer, coagulopathy, and smoking status, among others (see Supplementary material online, *Table S2*). Finally, we categorized patients based on their time to procedure (≤24 h or >24 h).

We used NRD hospital variables to determine hospital size based on the number of beds (small–medium/large), hospital teaching status, and hospital location (for teaching hospitals). We also estimated procedural volume by adding all admissions undergoing MT and CDT separately at each hospital. Hospitals were then categorized into quartiles according to procedural volume (first, second, third, and fourth quartiles), ensuring an equal number of each procedure across quartiles (*Table 1*). Due to the application of weight, slight differences in the sample across quartiles were observed.

Study outcomes

The primary outcome assessed was in-hospital mortality. Secondary in-hospital clinical outcomes included major bleeding, cardiac arrest, vascular complications, acute kidney injury (AKI) requiring dialysis, cardiopulmonary support, pericardial and respiratory complications. Major bleeding was defined as a composite of intracranial haemorrhage, blood transfusion requirement, gastrointestinal bleeding, and retroperitoneal bleeding. Cardiopulmonary support was a composite of vasopressor use, mechanical circulatory support and mechanical ventilation. Supplementary material online, *Table S3* lists the outcome definitions used. We studied resource utilization through length of stay (LOS), total cost of utilization in US dollars and use of cardiopulmonary support. Readmission outcomes included PE-related readmissions, post-discharge mortality during readmission, and all-cause readmissions. For 30-day, 6-, and 9-month outcomes, time-to-event was defined as the time from the index admission's discharge date to the event's occurrence date. In the case of multiple events, only the first readmission for particular outcome after the index admission was considered. The ICD-10 codes used to identify these outcomes are summarized in Supplementary material online, *Table S3*.

Finally, we calculated hospital costs by multiplying hospital charges with the corresponding cost-to-charge ratio.

Statistical analysis

Categorical variables were presented as numbers and percentages, and continuous variables as means with standard deviation (SD) or medians with interquartile range (IQR). χ^2 or Fisher exact tests were used to compare categorical variables between the two groups. The Student's *t*-test was used to compare continuous variables with normal distributions.[17]

Propensity score-matched cohorts for patients who underwent MT or CDT were generated through multivariable logistic regression, using all covariates described in Supplementary material online, *Table S1*. These covariates included patient demographics, comorbidities, and hospital characteristics like the location (in case of teaching hospital), and hospital bed size. Patients with similar propensity scores in the two groups were matched using a 1-to-1 scheme without replacement using a greedy method. A maximum propensity score difference (calliper width) of 0.1 was allowed between matched pair observations to keep the standardized difference < 10%. Patients without matched observations were excluded. Variables between two matched cohorts are depicted in *Table 1* and Supplementary material online, *Figure S1*. While calculating 30-day, 6-, and 9-month readmission outcomes, we excluded patients who died in the hospital as recommended by HCUP guidelines. We used a weighted sample as recommended by HCUP, with the potential for small differences in the sample size after 1:1 propensity-score matching.

Logistic regression was used to calculate the odds ratio (OR) with a 95% confidence interval (CI) for in-hospital outcomes in both the matched and unmatched cohorts. Further, Cox-proportional hazard regression was used to calculate the hazard ratio (HR) with a 95% CI for 30-day, 6-, and 9-month

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

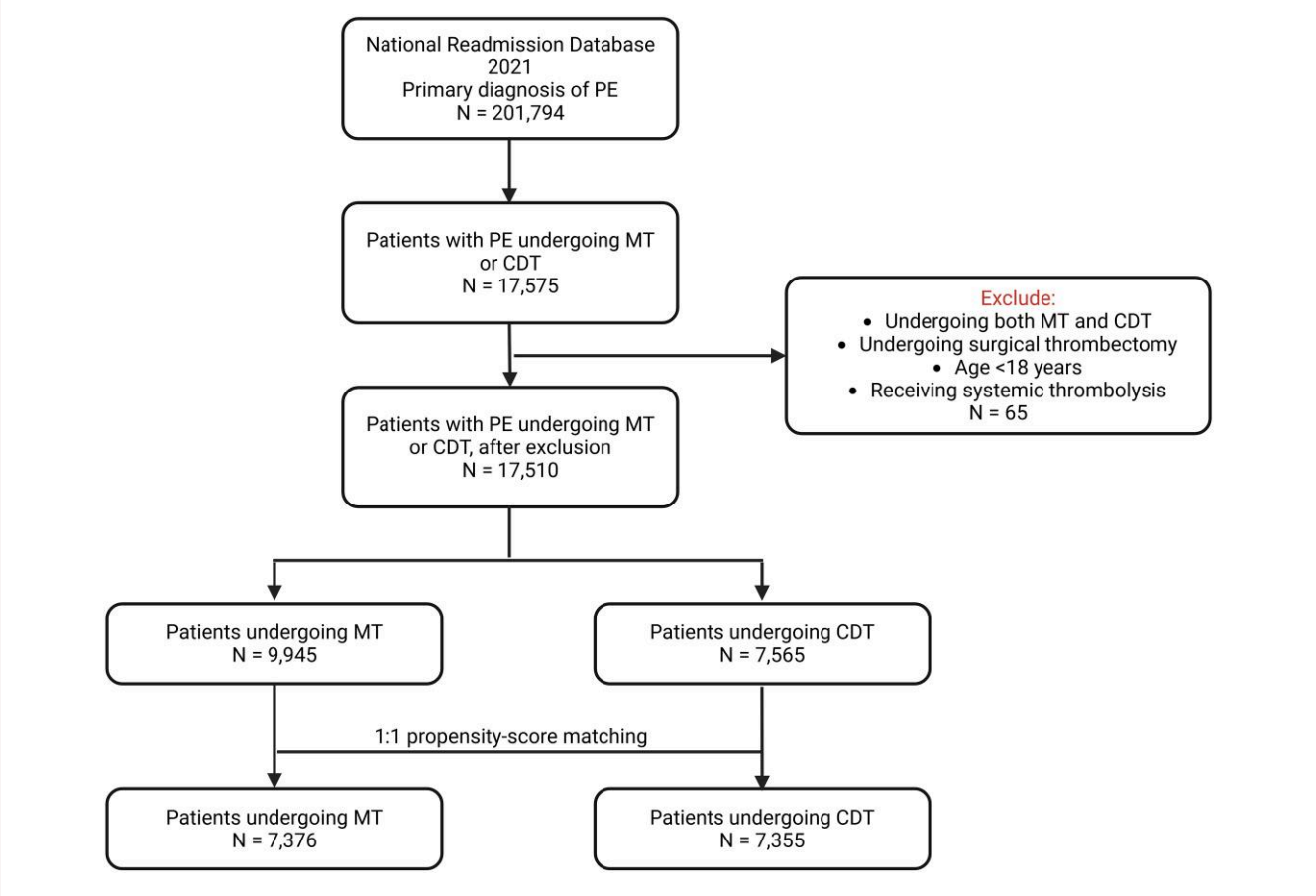


Figure 1 Patient selection flow diagram.

outcomes. Of note, the proportionality assumption was upheld for the Cox model across all outcomes, as indicated by the global test *P*-value being >0.05. Additionally, generalized linear model with a logit link and gamma distribution was used for LOS and cost, to compute the OR with a 95% CI. Hospital costs were obtained by multiplying the total hospital charges by relevant cost-to-charge ratio. Hospital costs reflected the total resource cost utilized during inpatient hospital stay for the index admission and include all procedure-related costs. For mortality analysis, a landmark analysis was conducted by distinguishing between in-hospital and post-discharge mortality during readmission. Hospital annual procedure volume of MT and CDT was analysed both as categorical and continuous variable with curvilinear (quadratic) regression analysis conducted to examine the association between annual procedure volume of MT and CDT group to the in-hospital mortality and major bleeding outcomes. Restricted cubic splines with random effect hierarchical models were created to assess the non-linear relationship between hospital's procedure volume and outcomes of in-hospital mortality and major bleeding as depicted in *Figure 2*.

Unmeasured bias analysis

Unmeasured confounders present the most significant challenge in retrospective cohort studies. Falsification endpoint and 'E-value' analyses were conducted to assess unmeasured confounders, based on prior studies.[18,19] For the falsification method, an alternative outcome, not expected to be directly influenced by the treatment being studied, was selected. Whether the study intervention (MT) affected these alternative outcomes was then evaluated using the same methodology applied to other study results. If no treatment effect is observed for the alternative outcome, it suggests, but does not conclusively prove, the potential for a causal treatment effect on the study outcomes. A combined measure of readmission due to gastrointestinal and urinary tract infections as the alternative outcome was chosen and the impact of the MT intervention on this measure was assessed. The E-value determines the minimum strength of association that unmeasured confounders would need to have with both the treatment and the outcome, given the measured and adjusted factors, to fully account for the observed association. This evaluates the potential relative risk required for any unmeasured confounding factor to negate the observed relationship between the study intervention (MT) and the study outcomes.

All *P*-values were assessed using a two-sided approach, with statistical significance established at $P < 0.05$. The statistical analyses were performed utilizing proper weighting, stratification and clustering of samples through the svy package within STATA, version 18.0 (StataCorp, TX).

Results

A total of 201 794 patients with a primary diagnosis of PE were identified. Of these patients, 9945 underwent endovascular MT and 7565 underwent CDT (*Figure 1*). After propensity-score matching, 7376 patients undergoing MT and 7355 patients undergoing endovascular CDT were analysed. For analysis of 30-day, 6-, and 9-month outcomes follow-up, matched pairs of 7055 patients who underwent MT and 7107 patients who underwent CDT were used.

Baseline characteristics

Table 1 shows baseline characteristics of patients in the MT and CDT groups before and after matching. Before matching, the mean age of both groups was similar (MT: 62.5 ± 14.8 vs. CDT: 61 ± 15.1; $P =$

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

Table 1 Baseline characteristics of endovascular MT vs. CDT before and after propensity-score matching

	Before matching			After matching		
	Endovascular MT	Endovascular CDT	*P*-value	Endovascular MT	Endovascular CDT	SMD (%)
	n = 9945	*n* = 7565		*n* = 7376	*n* = 7355	
Age (mean ± SD)	62.5 ± 14.8	61.5 ± 15.1	0.060	61.5 ± 15.2	61.7 ± 15.1	1.8
Female (%)	4742 (47.7%)	3713 (49.1%)	0.239	3622 (49.1%)	3555 (48.3%)	1.6
Comorbidities						
HTN	6374 (64.1%)	4814 (63.6%)	0.682	4638 (63.3%)	4689 (63.4%)	1.1
Diabetes	2821 (28.4%)	2016 (26.6%)	0.086	2003 (26.8%)	1968 (26.6%)	0.6
Hyperlipidaemia	4105 (41.3%)	3272 (43.2%)	0.114	3149 (42.2%)	3140 (42.1%)	0.1
Stroke/TIA	617 (6.2%)	326 (4.3%)	<0.001	314 (4.1%)	326 (4.1%)	1.5
CAD	2040 (20.5%)	1637 (21.6%)	0.283	1531 (20.4%)	1380 (21.3%)	2.6
CHF	1898 (19.1%)	1480 (19.6%)	0.683	1440 (19.2%)	1406 (19.5%)	0.3
Chronic lung disease	1907 (19.2%)	1623 (21.4%)	0.022	1554 (20.5%)	1520 (20.5%)	0.5
CKD	1182 (11.9%)	962 (12.7%)	0.291	930 (12.1%)	911 (12.3%)	0.6
Liver disease	591 (5.9%)	389 (5.1%)	0.158	383 (5.3%)	386 (5.4%)	2.1
Anaemia	557 (5.6%)	373 (4.9%)	0.207	425 (5.2%)	370 (5.1%)	0.7
Solid tumours or metastatic cancers or lymphoma	1070 (10.8%)	475 (6.3%)	<0.001	410 (6.0%)	474 (6.3%)	0.4
Smoking	2185 (21.9%)	1825 (24.1%)	0.038	1694 (22.9%)	1739 (23.3%)	0.3
Obesity	3901 (39.2%)	3392 (44.8%)	<0.001	3238 (42.8%)	3188 (43.1%)	0.9
Coagulopathy	1542 (15.5%)	1218 (16.1%)	0.508	1155 (16.2%)	1155 (15.8%)	1.5
Elixhauser comorbidity index, mean (SD)	4.5 (2)	4.4 (2)	0.274	4.4 (1.9)	4.4 (2.0)	0.5
Time to procedure						
≤24 h	8259 (83.1%)	6469 (85.5%)	<0.001	6107 (82.8%)	6303 (85.7%)	8.0
>24 h	1686 (16.9%)	1096 (14.5%)		1269 (17.2%)	1052 (14.3%)	
Bed size of the hospital[a]						
Small–medium	3405 (34.2)	3305 (43.7)	0.009	2954 (40.1%)	3099 (42.1%)	1.0
Large	6540 (64.5%)	4260 (56.3%)		4422 (59.9%)	4256 (57.9%)	
Teaching status of the hospital[b]						
Non-teaching	2107 (21.2%)	1782 (23.6%)	0.470	1695 (22.9%)	1663 (22.6%)	1.1
Teaching	7838 (78.8%)	5783 (76.4%)		5681 (77.0%)	5692 (77.4%)	
Hospital stratified by annual procedural volume[c]						
Quartile 1	2151 (21.6%)	1635 (21.6%)	0.920	1656 (22.5%)	1590 (21.6%)	1.7
Quartile 2	2323 (23.4%)	1728 (22.8%)		1721 (23.3%)	1666 (22.7%)	
Quartile 3	2374 (23.9%)	1976 (26.2%)		1762 (23.9%)	1930 (26.2%)	
Quartile 4	3097 (31.1%)	2226 (29.4%)		2238 (30.3%)	2169 (29.5%)	

CAD, coronary artery disease; CDT, catheter-directed thrombolysis; CHF, congestive heart failure; CKD, chronic kidney disease; HTN, hypertension; MT, mechanical thrombectomy; PAD, peripheral artery disease; TIA, transient ischaemic attack; SD, standard deviation; SMD, standard mean difference.

[a]The bedsize cut-off points are divided into small, median, and large have been done so approximately one-third of the hospitals in a given region, location, and teaching status combination would fall within each bed size category [Healthcare Cost and Utilization Project (HCUP) NRD Notes (ahrq.gov)].

[b]Teaching designation of the hospital is based on the Accreditation Council for Graduate Medical Education (ACGME) approved residency programme, Metropolitan hospitals were split into teaching and non-teaching status while nonmetropolitan cities were not based on the version of urban influence codes [Healthcare Cost and Utilization Project (HCUP) NRD Notes (ahrq.gov)].

[c]Number of hospitals and median [IQR] annual procedure volume in each quartile: MT—first (168 hospitals): 7.2 [4.5–9.6], second (64): 19.4 [16.7–23], third (33): 35.5 [31.4–41.9], fourth (17): 73.4 [55.2–112.6]; CDT—first (183): 5.6 [3.4–7.7], second (73): 14.4 [12.4–16.5], third (48): 30 [24.3–35.8], fourth (26): 60.6 [47.8–96.3]. Hospital number refers to post-matching group.

0.06), as was the proportion of female patients (MT: 47.7% vs. CDT: 49.1%; $P = 0.239$). Tumours (metastatic, solid tumours, and lymphoma) and history of stroke/ TIA were more common in the MT group compared with the CDT, while obesity, smoking, and chronic lung disease were more common in the CDT group compared with the MT group. Most patients in both groups (over 80%) underwent the

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

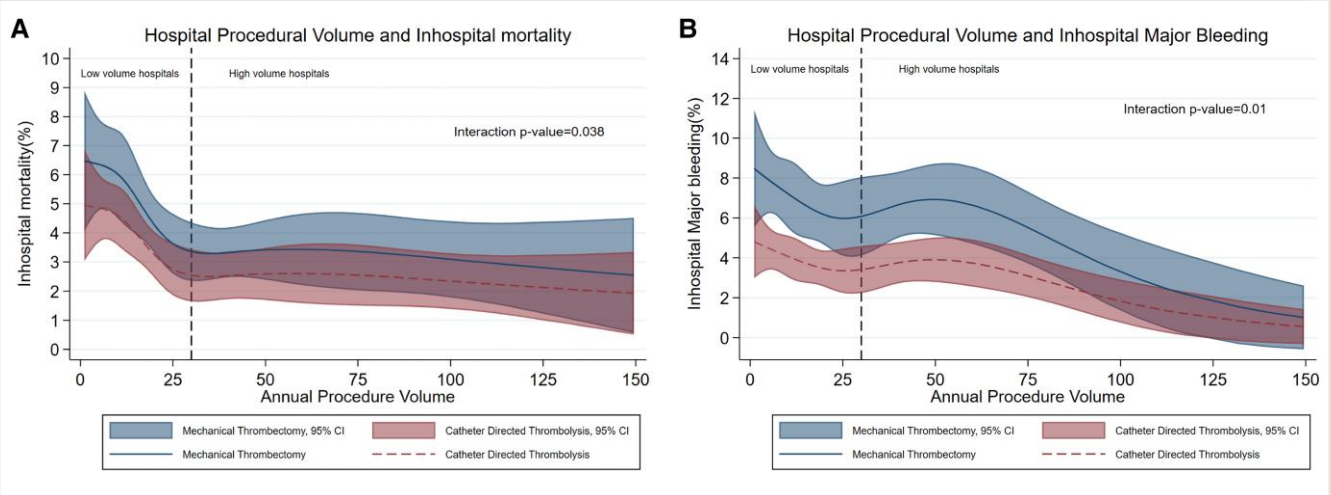


Figure 2 Spline graph for in-hospital mortality and major bleeding, comparing MT with CDT by annual procedure volume (interaction *P*-value for in-hospital mortality is 0.038, while that of in-hospital major bleeding is 0.01). The difference in the risk of in-hospital mortality and major bleeding between the two groups decreased as the procedure volume of the hospital increased. Black dotted line indicates cut-off between low-volume hospitals (first quartile; $n = 168$ hospitals; median 7.2 MT/year, 5.6 CDT/year) and high-volume hospitals (fourth quartile; $n = 26$; median 73.4 MT/year, 60.6 CDT/year).

procedure within 24 h of admission. A higher proportion of endovascular MT was done at large hospitals, but no difference was observed in the teaching status of hospital. *Table 1* also shows the baseline characteristics in the propensity score-matched cohort. The two groups were well-balanced, as presented graphically in Supplementary material online, *Figure S1*.

In-hospital outcomes

In the matched cohort, patients who underwent MT had higher in-hospital mortality compared with CDT (4.4% vs. 3.4%, respectively, OR: 1.31, 95% CI: 1.01–1.68; $P = 0.04$). Patients who underwent endovascular MT also had a higher incidence of cardiac arrest (2.8% vs. 1.2%, respectively, OR: 1.91, 95% CI: 1.33–2.74; $P < 0.001$), and major bleeding compared with the CDT group (6.3% vs. 3.6%, respectively, OR: 1.79, 95% CI: 1.39–2.32; $P < 0.001$). They also had a statistically significant, higher rate of respiratory complications, cardiopulmonary support and total cost of utilization compared with the CDT group (*Table 2*). In-hospital outcomes before propensity matching are described in Supplementary material online, *Table S4*.

These findings remained consistent even in sensitivity analyses. In patients with haemodynamically significant PE, patients undergoing MT had significantly greater in-hospital mortality after matching (2.2% vs. 1.4%, respectively, OR: 1.74, CI: 1.26–2.40, $P = 0.001$) (see Supplementary material online, *Table S5*). Among patients undergoing intervention < 24 hours since presentation, after matching, patients undergoing MT had greater in-hospital mortality, major bleeding, cardiac arrest, and total cost, as described in Supplementary material online, *Table S6*.

Short- and medium-term outcomes

There was no statistically significant difference between 30-day, 6-, and 9-month readmission outcomes (post-discharge mortality during readmission, number of PE readmissions), except for all-cause readmissions, which were higher in the MT group (*Table 3*). Readmission outcomes before propensity-score matching are detailed in Supplementary material online, *Table S7*.

Hospital procedure volume and outcomes

Higher procedural volume was associated with lower in-hospital mortality and major bleeding for both MT and CDT. Patients treated at high-volume centres (quartile 4; median [IQR] 73.4 [55.2–112.6] MT/year, 60.6 [47.8–96.3] CDT/year) had significantly better outcomes than those at low-volume centres (quartile 1; median [IQR] 7.2 [4.5–9.6] MT/year, 5.6 [3.4–7.7] CDT/year). In-hospital mortality was lower at high-volume centres (2.8% for MT and 2.2% for CDT) compared to low-volume centres (6.1% for MT and 4.8% for CDT; interaction $P = 0.038$; *Figure 2A*). Major bleeding followed a similar pattern (8.1% vs. 4.1% for MT and 4.9% vs. 2.5% for CDT; interaction $P = 0.01$). Notably, as hospital procedural volume increased, the difference in major bleeding risk between MT and CDT narrowed (*Figure 2B*).

Unmeasured confounder analysis

The falsification endpoint showed similar results between MT and CDT, suggesting that unmeasured confounders were likely balanced between the two groups. In the E-value analysis, the point estimate for mortality was higher for in-hospital mortality. This indicates that if an unmeasured confounder exists, it will need to have a stronger association with both the treatment and the outcome, even after accounting for variables used in propensity-score matching, to explain the observed effect. Therefore, it is unlikely that the current results were significantly affected by unmeasured confounders.

Discussion

CDT and MT have emerged as promising interventional options for intermediate- or high-risk PE. While smaller-scale and single-centre studies have demonstrated the safety and efficacy of these therapies, there remains a significant lack of large-scale real-world, head-to-head comparisons. In this comprehensive nationwide analysis, we compare MT and CDT, providing several important insights. First, MT was associated with higher rates of in-hospital mortality compared to CDT. Second, MT was also associated with higher rates of in-hospital major bleeding, along with greater costs, compared with CDT. Third,

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

Table 2 In-hospital outcomes in endovascular MT vs. endovascular CDT in propensity score-matched cohort[a]

	Endovascular MT ($n = 7376$)	Endovascular CDT ($n = 7355$)	Odds ratio	95% CI		*P*-value
Clinical outcomes						
In-hospital mortality	321 (4.4%)	248 (3.4%)	1.31	1.01	1.68	0.04
Cardiac arrest	204 (2.8%)	92 (1.2%)	1.91	1.33	2.74	<0.001
Vascular complications	53 (0.7%)	29 (0.4%)	1.85	0.94	3.66	0.075
AKI requiring dialysis	54 (0.7%)	30 (0.4%)	1.41	0.76	2.64	0.275
Major bleeding[b]	462 (6.3%)	264 (3.6%)	1.79	1.39	2.32	<0.001
Pericardial complications[c]	114 (1.6%)	87 (1.2%)	1.32	0.89	1.95	0.168
Respiratory complications[d]	344 (4.7%)	189 (2.6%)	1.45	1.12	1.87	0.004
Resource utilization						
Length of stay, median (IQR)	4 (2–6)	4 (3–6)	1.05	0.96	1.15	0.306
Total of cost of utilization ($), median (IQR)	28 167.8 (20 508.4–38 441.8)	24 321.6 (18 816.1–32 738.2)	1.68	1.53	1.84	<0.001
Cardiopulmonary support[e]	463 (6.3%)	252 (3.4%)	1.64	1.32	2.04	<0.001
E-value for mortality, point estimate, lower limit of CI		1.95, 1.11				

CAD, coronary artery disease; CHF, congestive heart failure; CI, confidence interval; CKD, chronic kidney disease; CDT, catheter-directed thrombolysis; IQR, interquartile range; MT, mechanical thrombectomy; TIA, transient ischaemic attack.

[a]Covariates used in propensity matching: age, gender, hypertension, diabetes mellitus, hyperlipidaemia, stroke/TIA, coronary artery disease, CHF, chronic liver disease, chronic renal failure, chronic lung disease, anaemia, all tumours (consisting of solid tumours, lymphoma and metastasis), anaemia, smoking, obesity, Elixhauser index, bed size of hospital, and teaching status of the hospital.

[b]Major bleeding—intracranial haemorrhage, GI bleed, blood transfusion, and retroperitoneal bleeding.

[c]Pericardial complications—pericardiocentesis, haemopericardium, pericardial effusion.

[d]Respiratory complications—respiratory complications with or without need of mechanical ventilation.

[e]Cardiopulmonary support—vasopressor use, mechanical circulatory support, and mechanical ventilation.

importantly, in-hospital mortality and major bleeding for both MT and CDT, as well as the difference in outcomes between MT and CDT, reduced with greater procedural volumes (*Graphical Abstract*).

MT being linked to higher in-hospital mortality than CDT is noteworthy, and likely multifactorial. One possible hypothesis is that thrombus fragmentation during MT might cause distal embolization into the distal pulmonary vasculature. Further, higher major bleeding rates with MT may have contributed to greater mortality. The rates of in-hospital mortality with CDT in our cohort align with those of US-CDT in the SEATTLE II study.[7] Contrastingly, the mortality rate for MT in our study is significantly higher than that reported in previously published studies such as FLARE, FLASH, and EXTRACT-PE.[8,10,20] PEERLESS found much lower rates of 30-day mortality in both groups, without significant difference between the two arms.[12] There are multiple possible reasons for these differences. Unlike prior studies, which focused on intermediate-risk patients, PE patients in our sample were not risk-stratified, so a broader spectrum of patients, including both intermediate- and high-risk patients, was included. Even though we performed meticulous propensity-score matching, we lacked access to details of clinical presentation, laboratory, or imaging data. So, it is conceivable that MT patients were at inherently higher risk than CDT patients despite our statistical adjustment. This is suggested by higher baseline cancer and stroke, along with greater rates of cardiac arrest in the MT cohort. Alternatively, it is possible that the outcomes of these trials were reflective of the high procedural expertise inherent to tertiary referral centres, along with the more stringent patient selection criteria. Reporting bias may have limited direct comparisons. PEERLESS utilized a hierarchical win ratio composite of all-cause mortality, intracranial haemorrhage, major bleeding, clinical deterioration or escalation to bailout, and postprocedural intensive care unit admission and length of stay. While this favoured MT over CDT, this choice of primary outcome may not fully align with day-to-day clinical decision-making. Thus, our findings, with a hard endpoint of all-cause mortality, complement those of PEERLESS.

In a recent meta-analysis, Ismayl *et al.* identified eight studies (totalling 1400 patients) that compared outcomes of CDT and thrombectomy (both MT and surgical thrombectomy). They found a lower mortality rate with CDT compared with thrombectomy overall, but no significant difference between CDT and MT alone. Notably, this meta-analysis was limited by the inclusion of small, mostly single-centre, observational studies.[15] In a more recent meta-analysis comparing MT and CDT, using goals from prior historical studies, Zhang *et al.* reported no significant differences between the two in terms of mortality, but reported that MT was superior in terms of major bleeding.[21] In the REAL-PE study, investigators utilized a 16-health system electronic medical record-based repository (Truveta Inc.) to identify patients with PE who underwent CDT or MT (using the device from Inari Medical). In their contemporary analysis (2018–2023) of 1137 CDT and 661 MT patients, while they found that MT was associated with a higher rate of major bleeding, they did not note an association with in-hospital mortality, LOS, or 30-day all-cause readmissions. No propensity match was performed, and there were significant baseline differences between groups.[22] In PEERLESS, the investigators did not find significant differences in major bleeding between the two groups.[12]

In our current analysis, MT was associated with a greater risk of major bleeding than CDT, with rates of 6.3% and 3.6%, respectively. In prior studies evaluating US-CDT, the SEATTLE II study reported a major bleeding rate of 10%,[7] the KNOCOUT PE study found a rate of 1.6%,[23] and the OPTALYSE PE trial,[24] evaluating different regimens of thrombolytics in US-CDT, found a rate of 4%. Among studies of MT, the FLAME study,[25] which evaluated the FlowTriever MT system in high-risk PE, reported a major bleeding rate of 11.5%. In contrast,

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

Table 3 Post-discharge 30-day, 6-month, and 9-month outcomes in endovascular MT vs. endovascular CDT in the propensity score-matched cohort

	Endovascular MT ($n = 7055$)	Endovascular CDT ($n = 7107$)
Post-discharge mortality		
No. of patients with events	75	67
Cumulative event rate, %		
At 30 days	0.9	0.7
At 6 months	1	0.9
At 9 months	1.1	0.9
HR (95% CI; *P*-value)	1.23 (0.77–1.99; 0.380)	
Falsification outcome[a]	1.05 (0.61–1.84; 0.840)	
PE readmissions		
Number of PE readmissions	135	106
Cumulative event rate, %		
At 30 days	1.6	1.3
At 6 months	1.9	1.5
At 9 months	1.9	1.5
HR (95% CI; *P*-value)	1.37 (0.94–1.98; 0.097)	
All-cause readmissions		
Number of all-cause readmissions	1051	1093
Cumulative event rate, %		
At 30 days	7.1	6.4
At 6 months	10.5	10.6
At 9 months	14.9	15.4
HR (95% CI; *P*-value)	1.18 (1.04–1.34; 0.013)	

CDT, catheter-directed thrombolysis; CI, confidence interval; HR, hazard ratio; MT, mechanical thrombectomy; PE, pulmonary embolism.
[a]Falsified endpoint is the composite of gastrointestinal or urinary tract infection.

the FLASH registry,[8] focusing on MT in intermediate-risk PE, observed a lower major bleeding rate of around 1.4%, while the EXTRACT-PE study of the Indigo MT system reported a rate of 1.7%, with half of its cohort being intermediate-risk.[20] However, comparison is limited by the fact that these studies utilized different definitions of major bleeding and had a high predominance of intermediate-risk patients and selected centres. The increased bleeding risk with MT could be attributed to the use of larger bore catheters (16–24 F) that potentially cause more access site injury as well as trauma to the pulmonary vasculature, and the need for advanced expertise and procedural experience. In contrast, CDT utilizes smaller catheters (typically 4–8 F) to deliver thrombolytics locally, leading to less risk of procedure-related vascular complications. MT is often performed in patients considered to be at a higher bleeding risk or with contraindications to the use of thrombolytics, although evidence that this results in less bleeding is lacking. Our real-world data challenges the assumption that MT results in less bleeding. Overall, our findings suggest that patient, device selection, physician, and centre experience all potentially play a role in determining outcomes.

We found that hospitals with higher annual procedural volumes experienced lower in-hospital mortality and lower rates of major bleeding for both MT and CDT. This suggests that procedural outcomes improve with experience and higher centre procedural volumes. This has been previously observed in other cardiovascular interventions, such as coronary interventions and MT for ischaemic stroke.[26,27] We also found that the difference in major bleeding rates between MT and CDT reduced at higher volume centres. This narrowing gap in major bleeding rates between MT and CDT at high-volume centres indicates that MT outcomes may be more sensitive to operator and centre experience. Further, it may emphasize the potential benefit of centralizing care for complex PE interventions at high-volume centres, perhaps utilizing a hub-and-spoke model to ensure optimal patient outcomes. Another potential explanation for this finding is differing patient selection for procedural care between high- and low-volume centres—low-volume centres may reserve MT solely for higher-risk patients.

Previous single-arm studies and registries, such as FLARE, FLASH, and SEATTLE II, have provided insights into the safety and efficacy of MT and CDT but lacked direct comparisons between the two modalities.[7–10] The recently published PEERLESS randomized clinical trial filled that void by directly comparing both treatment modalities. They found no significant difference in 30-day mortality between the two treatment arms and reported similar rates of major bleeding across groups. However, PEERLESS showed a shorter length of stay (LOS) and fewer 30-day readmissions in the MT group compared to the CDT group. The 30-day mortality rate observed in PEERLESS was notably lower than that seen in real-world analyses, likely reflecting its focus on an intermediate-risk population.[12] Our study, in contrast, consists of a broader mix of intermediate- and high-risk patients, which may have contributed to the higher mortality rates compared to PEERLESS. While major bleeding rates in the MT group were similar between the two studies, the CDT group in PEERLESS reported a higher major bleeding rate (6.9%) than in our analysis (3.6%), despite applying stricter ISTH criteria.[12] Notably, our analysis of hospital volume and outcomes revealed that the differences in all-cause mortality and major bleeding narrowed between two arms, and the absolute rates of both outcomes decreased, in high-volume centres—findings that align with PEERLESS where all procedures are performed at tertiary centre and by the experienced physicians. Moreover, our study included all types of device in both arms, whereas PEERLESS was focused on Flowtriever for MT and EKOS, Cragg-McNamara™, Uni-Fuse™. While PEERLESS, as the first RCT in this field, has provided important insights into these therapies,[12] our study adds valuable real-world outcomes from a large, propensity-matched comparison, supplemented by additional statistical methods to account for unmeasured bias.

While our study provides valuable insights, it has several limitations, including its retrospective design and reliance on administrative coding, which may have led to potential misclassification biases. The CDT group did not differentiate between standard CDT and ultrasound-assisted CDT, while the MT group included all devices available in the USA at the time of the study without identifying specific device types or bore sizes. Finally, the lack of clinical presentation data and imaging data did not allow us to risk-stratify our sample. So, potential unmeasured imbalances between the groups with respect to those characteristics could not be corrected with propensity matching. This may have affected study results. However, these concerns are mitigated by our falsification endpoint and E-value analyses that showed that our results were unlikely to be significantly affected by unmeasured confounders.

Conclusions

In our nationwide, real-world study of patients with PE treated with endovascular therapy, we found that CDT was associated with lower in-hospital mortality and major bleeding compared to MT after propensity-score matching. In-hospital mortality and major bleeding for both MT and CDT reduced with increased hospital procedural volume. These findings highlight potential promise of CDT, as well as the potential need for a hub-and-spoke model for PE care, warranting further prospective studies.

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026

Supplementary material

Supplementary material is available at *European Heart Journal: Acute Cardiovascular Care* online.

Funding

None.

Conflict of interest: P.M. is an advisory board member for Abbott, Boston Scientific, Medtronic, and RapidAI. He is a consultant for Abbott, Penumbra, Boston Scientific, and Medtronic. He reports speaker honoraria from Boston Scientific and Medtronic. None of the other authors have any conflicts of interest or relevant disclosures.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

References

1. Tsao CW, Aday AW, Almarzooq ZI, Alonso A, Beaton AZ, Bittencourt MS, *et al.* Heart disease and stroke statistics—2022 update: a report from the American Heart Association. *Circulation* 2022;**145**:e153–e639.
2. Pauley E, Orgel R, Rossi JS, Strassle PD. Age-stratified national trends in pulmonary embolism admissions. *Chest* 2019;**156**:733–742.
3. Konstantinides SV, Meyer G, Becattini C, Bueno H, Geersing G-J, Harjola V-P, *et al.* 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). *Eur Heart J* 2020;**41**: 543–603.
4. Pruszczyk P, Klok FA, Kucher N, Roik M, Meneveau N, Sharp AS, *et al.* Percutaneous treatment options for acute pulmonary embolism: a clinical consensus statement by the ESC Working Group on Pulmonary Circulation and Right Ventricular Function and the European Association of Percutaneous Cardiovascular Interventions. *EuroIntervention* 2022;**18**:e623–e638.
5. Marti C, John G, Konstantinides S, Combescure C, Sanchez O, Lankeit M, *et al.* Systemic thrombolytic therapy for acute pulmonary embolism: a systematic review and meta-analysis. *Eur Heart J* 2015;**36**:605–614.
6. Kucher N, Boekstegers P, Müller OJ, Kupatt C, Beyer-Westendorf J, Heitzer T, *et al.* Randomized, controlled trial of ultrasound-assisted catheter-directed thrombolysis for acute intermediate-risk pulmonary embolism. *Circulation* 2014;**129**:479–486.
7. Piazza G, Hohlfelder B, Jaff MR, Ouriel K, Engelhardt TC, Sterling KM, *et al.* A prospective, single-arm, multicenter trial of ultrasound-facilitated, catheter-directed, low-dose fibrinolysis for acute massive and submassive pulmonary embolism: the SEATTLE II study. *JACC Cardiovasc Interv* 2015;**8**:1382–1392.
8. Toma C, Jaber WA, Weinberg MD, Bunte MC, Khandhar S, Stegman B, *et al.* Acute outcomes for the full US cohort of the FLASH mechanical thrombectomy registry in pulmonary embolism. *EuroIntervention* 2023;**18**:1201–1212.
9. Sardar P, Piazza G, Goldhaber SZ, Liu P-Y, Prabhu W, Soukas P, *et al.* Predictors of treatment response following ultrasound-facilitated catheter-directed thrombolysis for submassive and massive pulmonary embolism: a SEATTLE II substudy. *Circ Cardiovasc Interv* 2020;**13**:e008747.
10. Tu T, Toma C, Tapson VF, Adams C, Jaber WA, Silver M, *et al.* A prospective, single-arm, multicenter trial of catheter-directed mechanical thrombectomy for intermediate-risk acute pulmonary embolism: the FLARE study. *JACC Cardiovasc Interv* 2019;**12**:859–869.
11. Horowitz JM, Jaber WA, Stegman B, Rosenberg M, Fanola C, Bhat AP, *et al.* Mechanical thrombectomy for high-risk pulmonary embolism: insights from the US cohort of the FLASH registry. *J Soc Cardiovasc Angiogr Interv* 2023;**3**:101124.
12. Jaber WA, Gonsalves CF, Stortecky S, Horr S, Pappas O, Gandhi RT, *et al.* Large-bore mechanical thrombectomy versus catheter-directed thrombolysis in the management of intermediate-risk pulmonary embolism: primary results of the PEERLESS randomized controlled trial. *Circulation* 2025;**151**:260–273.
13. HCUP Nationwide Readmissions Database (NRD), Healthcare cost and Utilization Project (HCUP). 2016, 2017, 2018, and 2019. Agency for Healthcare Research and Quality, Rockville, MD. Available at: www. hcup-us.ahrq.gov/. Accessed September 11, 2024.
14. Sedhom R, Beshai R, Moussa P, Megaly M, Mohsen A, Abramov D, *et al.* Outcomes with malignancy-associated high-risk pulmonary embolism: a nationwide analysis. *Mayo Clin Proc* 2024;**99**:81–89.
15. Ismayl M, Ismayl A, Hamadi D, Aboeata A, Goldsweig AM. Catheter-directed thrombolysis versus thrombectomy for submassive and massive pulmonary embolism: a systematic review and meta-analysis. *Cardiovasc Revasc Med* 2024;**60**:43–52.
16. Moore BJ, White S, Washington R, Coenen N, Elixhauser A. Identifying increased risk of readmission and in-hospital mortality using hospital administrative data: the AHRQ Elixhauser comorbidity index. *Med Care* 2017;**55**:698–705.
17. Khera R, Angraal S, Couch T, Welsh JW, Nallamothu BK, Girotra S, *et al.* Adherence to methodological standards in research using the national inpatient sample. *JAMA* 2017; **318**:2011–2018.
18. Majmundar M, Patel KN, Doshi R, Anantha-Narayanan M, Kumar A, Reed GW, *et al.* Comparison of 6-month outcomes of endovascular vs surgical revascularization for patients with critical limb ischemia. *JAMA Netw Open* 2022;**5**:e2227746.
19. Majmundar M, Patel KN, Doshi R, Mehta H, Vindhyal MR, Hance KA, *et al.* Prognostic value of hospital frailty risk score and clinical outcomes in patients undergoing revascularization for critical limb-threatening ischemia. *J Am Heart Assoc* 2023;**12**:e030294.
20. Sista AK, Horowitz JM, Tapson VF, Rosenberg M, Elder MD, Schiro BJ, *et al.* Indigo aspiration system for treatment of pulmonary embolism: results of the EXTRACT-PE trial. *JACC Cardiovasc Interv* 2021;**14**:319–329.
21. Zhang RS, Maqsood MH, Greco AA, Postelnicu R, Mukherjee V, Alviar CL, *et al.* Efficacy and safety of catheter-based thrombectomy versus catheter-directed thrombolysis in acute pulmonary embolism. *Am Heart J* 2024;**272**:109–112.
22. Monteleone P, Ahern R, Banerjee S, Desai KR, Kadian-Dodov D, Webber E, *et al.* Modern treatment of pulmonary embolism (USCDT vs MT): results from a real-world, big data analysis (REAL-PE). *J Soc Cardiovasc Angiogr Interv* 2023;**3**:101192.
23. Sterling KM, Goldhaber SZ, Sharp ASP, Kucher N, Jones N, Maholic R, *et al.* Prospective multicenter international registry of ultrasound-facilitated catheter-directed thrombolysis in intermediate–high and high-risk pulmonary embolism (KNOCOUT PE). *Circ Cardiovasc Interv* 2024;**17**:e013448.
24. Tapson VF, Sterling K, Jones N, Elder M, Tripathy U, Brower J, *et al.* A randomized trial of the optimum duration of acoustic pulse thrombolysis procedure in acute intermediate-risk pulmonary embolism: the OPTALYSE PE trial. *JACC Cardiovasc Interv* 2018;**11**:1401–1410.
25. Silver MJ, Gibson CM, Giri J, Khandhar S, Jaber W, Toma C, *et al.* Outcomes in high-risk pulmonary embolism patients undergoing FlowTriever mechanical thrombectomy or other contemporary therapies: results from the FLAME study. *Circ Cardiovasc Interv* 2023;**16**:e013406.
26. Saito Y, Inohara T, Kohsaka S, Ando H, Ishii H, Yamaji K, *et al.* Volume-outcome relations of percutaneous coronary intervention in patients presenting with acute myocardial infarction (from the J-PCI registry). *Am J Cardiol* 2023;**192**:182–189.
27. Stein LK, Mocco J, Fifi J, Jette N, Tuhrim S, Dhamoon MS. Correlations between physician and hospital stroke thrombectomy volumes and outcomes: a nationwide analysis. *Stroke* 2021;**52**:2858–2865.

Downloaded from https://academic.oup.com/ehjacc/article/14/12/711/8240312 by IT Services at Oxford University - James Cummings user on 11 February 2026